

Exercise 2.20. The Pythagorean theorem involves integers a , b and c for which $a^2 + b^2 = c^2$. Prove that if three integers satisfy this relationship, then either a or b will be divisible by 3.

Solution.

We are given three integers a, b, c satisfying the Pythagorean equation

$$a^2 + b^2 = c^2.$$

We want to prove that at least one of a or b is divisible by 3.

Step 1: Consider the possible remainders modulo 3

Any integer has one of these forms:

$$n \equiv 0, 1, \text{ or } 2 \pmod{3}.$$

Squaring them gives:

$$0^2 \equiv 0 \pmod{3},$$

$$1^2 \equiv 1 \pmod{3},$$

$$2^2 = 4 \equiv 1 \pmod{3}.$$

Therefore, every integer squared is either 0 or 1 $\pmod{3}$.

So,

$$a^2, b^2, c^2 \equiv 0 \text{ or } 1 \pmod{3}.$$

Step 2: Suppose neither a nor b is divisible by 3

If $3 \nmid a$, then

$$a^2 \equiv 1 \pmod{3}.$$

Likewise, if $3 \nmid b$,

$$b^2 \equiv 1 \pmod{3}.$$

Therefore,

$$a^2 + b^2 \equiv 1 + 1 \equiv 2 \pmod{3}.$$

But we know

$$a^2 + b^2 = c^2.$$

Thus,

$$c^2 \equiv 2 \pmod{3}.$$

Step 3: Contradiction

But we established that the square of any integer modulo 3 can only be

$$0 \text{ or } 1.$$

It can never be 2.

Therefore,

$$c^2 \not\equiv 2 \pmod{3},$$

which is a contradiction.

Hence our assumption that neither a nor b is divisible by 3 must be false.

Therefore,

$$\boxed{3 \mid a \text{ or } 3 \mid b}.$$

The core idea

The entire proof rests on the fact that

$$\boxed{n^2 \equiv 0 \text{ or } 1 \pmod{3}}.$$

If both a and b were not divisible by 3, then

$$a^2 + b^2 \equiv 1 + 1 \equiv 2 \pmod{3},$$

but a square can never be $2 \pmod{3}$.